Wilson Abdón Ordóñez Ramos

Data Scientist in banking · Master's in Data Science, PUC Chile

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Summary

Data scientist at Banco Guayaquil, where I have worked in retail banking analytics since 2018. I build models for churn and retention, customer lifetime value (CLV), next best offer, transaction network analysis, and natural language processing with embeddings. I work daily with Python, R, SQL, PySpark, and Databricks. Master's in Data Science from Pontificia Universidad Católica de Chile (2026), engineering degree in Statistics and Computing from ESPOL, and co-author of a paper in *ACS Omega*. Before banking, I was a teaching technician and research assistant at ESPOL.

Professional experience

Banco Guayaquil S.A. `Nov 2018 – present`
Guayaquil, Ecuador · Data Analytics, Retail Banking Tribe

Data Scientist `Jan 2025 – present`
Senior Officer `Apr 2023 – Dec 2024`
Officer `Nov 2018 – Mar 2023`

- Statistical analysis and models to support business decisions and understand customers' financial behavior.
- Churn and product-adoption models, spending profiles, and transaction network analysis; data collection through web scraping and APIs.
- Contributed to automating processes in the Risk department.

Projects at Banco Guayaquil

Primary Bank Model 2025
PySpark · Databricks · Hive · XGBoost · NetworkX · igraph

Designed and built a model that identifies customers who use the bank as their primary financial institution, combining factor analysis, XGBoost, and feature engineering on transaction data. Includes centrality analysis of money-transfer networks.

Sentiment analysis of customer communications 2025
Python · R · OpenAI API · PCA · ICA

Built a natural language processing pipeline on customer messages, using OpenAI embeddings and dimensionality reduction with PCA and ICA, to classify messages and detect patterns.

Return on Experience and CSAT sensitivity 2025
Econometrics

Developed econometric models that measure how profitability (ROA, ROE) responds to CSAT and NPS: a framework to estimate the return on digital experience improvements.

Customer Lifetime Value (CLV) 2023
R · survival · earth · XGBoost · Sparklyr

Built a customer value estimation system from deposits and loans: survival analysis to estimate customer lifetime and regression models to project financial behavior.

Next Best Offer from product adoption paths 2023
Recommender systems

Modeled recommendations from product adoption patterns and acquisition sequences, for cross-selling and retention campaigns.

Multi-product churn 2022
R · GLMNET · XGBoost · Sparklyr

Predicted churn in insurance, deposits, and other products with GLMNET and XGBoost; built spending profiles and customer segments.

Other business analyses 2018 – present

Forecasting corporate deposits from transaction data and graph analysis. Demand and sustainability analysis for Bancos de Barrio, the bank's neighborhood banking agents.

Generative AI and assisted development

- AI-assisted development with Claude and Codex: requirements specification and iterative code review for data analysis and automation.
- Structured prompt design with defined objectives, constraints, acceptance criteria, and deliverables.
- Agent-based workflows for data science; working knowledge of reusable skills and the MCP protocol.
- AI-assisted preparation of visualizations, reports, presentations, and technical documents in LaTeX.

Research and teaching

ESPOL — Research Assistant May 15 – Oct 31, 2021
Escuela Superior Politécnica del Litoral, Guayaquil

Methodology, validation, and formal analysis for the study published in *ACS Omega* (2023), listed under Publication.

ESPOL — Teaching Technician Jul 2017 – Sep 2018
Faculty of Natural Sciences and Mathematics

- Teaching support and pre/post analysis of physics concept inventories: rotation and balance (RMSS) and force (FCI).
- Forecasting of student demand and course pass rates with XGBoost and time series.
- Analysis of the natural disaster program at AJA park; administrative tools for the faculty office.

Early experience

Hospital León Becerra — Intern Aug – Sep 2016
Guayaquil, Ecuador

Supported a statistical study of patient admissions.

ESPOL — Statistics Assistant, Graduate School 2014 – 2016
Faculty of Natural Sciences and Mathematics

Market and demand studies to launch graduate programs in Mathematics, Education, Environmental Sciences, Applied Chemistry, and Physics Teaching.

INAMHI — Research Intern Apr 7 – May 8, 2015
National Institute of Meteorology and Hydrology · 200 hours

Monthly rainfall analysis and ROC curve programming.

ESPOL — Assistant at CEIE 2014
Center for Statistical Studies and Research · 200 certified hours

Data entry, consistency checks, and tabulation; statistical reports and time-series analysis of crime in Guayaquil.

Education

Master's in Data Science conferred Apr 10, 2026
Pontificia Universidad Católica de Chile · Magíster en Ciencia de Datos

Big data architecture and machine learning: Spark, Hadoop, MapReduce, Bayesian methods, deep learning, hypothesis testing, AI ethics, and model explainability.

Engineering degree in Statistics and Computing 2012 – 2017
Escuela Superior Politécnica del Litoral (ESPOL), Ecuador · Ingeniero en Estadística Informática

Conferred May 29, 2017. Registration no. 1021-2017-1855694.

Thesis (in Spanish) on Ecuador's income distribution with robust quantile regression: *Análisis de la estructura en la distribución de los ingresos de los ecuatorianos usando modelo de regresión cuantílica robusta* (ESPOL repository).

High school diploma, physics and mathematics track 2012
Unidad Educativa San José La Salle, Guayaquil

Publication

Bustos, D.; Galarza, C.; **Ordoñez, W.**; Brauchi, S.; Benso, B. (2023). Cost-Effective Pipeline for a Rational Design and Selection of Capsaicin Analogues Targeting TRPV1 Channels. *ACS Omega*, 8(13), 11736–11749. Published online March 24, 2023.

doi.org/10.1021/acsomega.2c05672 · full text on PubMed Central

Contribution (CRediT): methodology, validation, formal analysis, review and editing. Affiliation: FCNM, ESPOL.

Technical skills

Programming: Python and R (advanced); SQL.

Big data: Apache Spark, PySpark, Sparklyr, SparkR, Databricks, Hive, Hadoop, MapReduce.

Modeling: XGBoost, GLMNET, regression and quantile regression, survival analysis, time series, segmentation, factor analysis, graph theory.

NLP and analytics: Embeddings, OpenAI API, PCA, ICA, sentiment analysis.

Libraries: survival, earth, igraph, NetworkX, pandas, scikit-learn.

Visualization: Power BI, Tableau, R Shiny, ggplot2, Matplotlib, Plotly.

Databases and pipelines: SQL Server, PostgreSQL, MySQL, SQLite, MongoDB, ETL processes, Git.

Other tools: SPSS, Minitab, Wolfram Mathematica, LaTeX, Oracle BI, advanced Excel and Access.

Languages

Spanish (native) · English (B1, Common European Framework of Reference).

Centro Ecuatoriano Norteamericano, 2009–2011; ELASH certificate.

Certifications and courses

SCRUM Master Professional Certificate (SMPC) Feb 2019
Certiprof · credential 71551126611886

Business Intelligence with Tableau · Ecuadorian Statistical Society Sep 2017
Cartographic Updating · ESPOL · 10 hours Oct 2016
Introduction to Stochastic Processes · ESPOL · 10 hours Oct 2016
Redatam+SP as a data dissemination tool · ESPOL · 10 hours Oct 2016
Introduction to data management with Infostat and R · ESPOL Nov 2014
Statistical Pattern Recognition with Classification Trees · ESPOL Oct 2014

Academic activities

19th Statistics and Computing Conference Oct 2016
ESPOL
Talk: *Multivariate analysis of relationships between open-ended text and categorical or numeric variables: an application to crime statistics.*

18th Statistics Conference Oct 2014
ESPOL · 20 hours

References

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